Interpreting Imagined Futures: Discovering How World Action Models Steer Robot Behavior

Anonymous Author(s)
Affiliation
Address
email

Abstract

World action models produce predicted futures alongside robot actions, but co-generation alone does not causally establish how the predicted future affects the action. Prior work shows that behavior depends on information at predicted-future token positions, but not how particular future content shapes the resulting action. We use two intervention stages to characterize that influence: Stage 1 measures how strongly and in what direction future content changes behavior, and Stage 2 tests how much of that effect is carried through future-token attention keys and values. Stage 1 starts with two normal rollouts from the same simulator state. We insert the future representation associated with one rollout (the *donor*)—either its predicted future or an encoding of its executed continuation—into the other (the *recipient*), hold all other inputs fixed, and measure whether the recomputed action moves toward the donor action, which the model never sees. Across 22 states and six RoboLab tasks, donor predicted futures redirect Cosmos 3 actions and executed endpoints almost completely toward the donor rollout. Cosmos Policy shows a smaller effect at the first decision state of all ten LIBERO-Long tasks. Matched controls rule out generic perturbation and replacement as complete explanations. Content interventions localize Cosmos Policy's effect to visual future targets, especially the wrist camera, whereas isolated robot or object pixels do not reproduce Cosmos 3's complete-video effect. For Cosmos 3, we keep the donor's predicted future inserted but restore the recipient's attention keys and values at predicted-future token positions. This removes 83–88% of mean steering, with a positive reduction in all 22 tested states, showing that this interface carries most of the measured effect.

1 Introduction

Many world action models (WAMs) produce predicted futures alongside robot actions [8, 14]. Joint generation, however, does not imply that action computation uses the predicted future: a policy could map the current observation and instruction directly to an action while producing a plausible future as an auxiliary output. Alternatively, predicted-future representations may serve as a computational workspace, encode a prospective robot pose for inverse dynamics [1, 7], or represent consequences used to evaluate actions [2, 3].

RIFT provides the closest prior causal evidence by masking reads from and editing values at predicted-future positions in several WAMs [22]. These interventions establish that behavior depends on information carried through the predicted-future interface. Because disrupting the interface can move an action in any direction, however, they do not reveal how a particular coherent future shapes behavior. We adapt mechanistic interventions to this joint video–action modality to make directional, testable claims about both future content and its computational pathway.

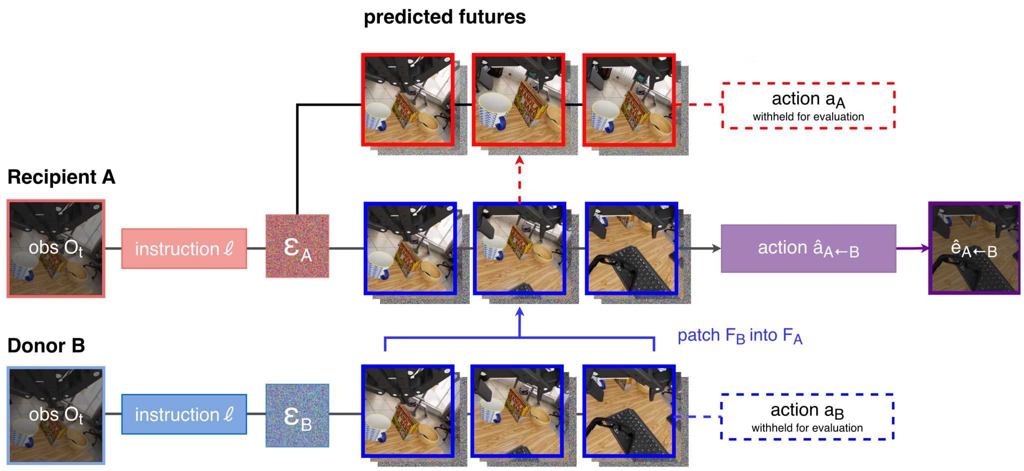

Figure 1: Future-content intervention. From two native rollouts at the same state, Stage 1 inserts donor B's future target into recipient A at every denoising step while holding the current observation fixed. We test whether the recomputed action and endpoint move toward B; the donor action is withheld and used only for evaluation. Cosmos Policy uses an executed-endpoint encoding; Cosmos 3 tests predicted latents and executed videos.

We ask three questions: Does coherent future content direct behavior; what visual content explains the effect; and which internal pathway carries it? In Stage 1, we insert one normal rollout's future representation into another rollout from the same saved state. The receiving rollout is the *recipient*; the source is the *donor*. We recompute the recipient's action while holding all other inputs fixed. The donor action is never shown to the model and serves only as a directional reference after inference. For Cosmos 3, Stage 2 keeps the donor future inserted but restores keys and values at predicted-future positions from the recipient's native run, testing how much of the Stage-1 effect passes through that interface.

The experiments yield three findings. First, Cosmos 3 donor futures redirect actions and executed endpoints almost completely toward the donor across 22 states and six tasks; Cosmos Policy shows a smaller early-state effect across all ten LIBERO-Long tasks. Second, Cosmos Policy's effect is concentrated in visual future targets, especially the wrist camera, whereas isolated robot or object pixels cannot reproduce Cosmos 3's complete-video effect. Third, restoring recipient K/V at predicted-future positions removes 83–88% of mean Cosmos 3 steering, with a positive reduction in all 22 tested states. Together, the interventions support two concrete claims: coherent donor futures predict the direction of action changes, and future-token K/V carries most of that change in Cosmos 3.

Related work. Visual robot policies may infer actions from predicted video, condition on its representation, or generate futures and controls jointly [1, 6–8, 10, 14, 20]. These objectives do not establish how a particular future participates in inference. Recent work tests whether explicit rollout is needed [10, 21], aligns foresight and action representations [17], or intervenes on predicted-future positions [22]. Activation and path patching test whether a representation or route affects output [5, 12, 18]. We use coherent within-state counterfactuals and a directional metric to identify the effect carried through predicted-future K/V [23].

2 Two-Stage Intervention Design

2.1 Stage 1: Future-content intervention

Starting from saved simulator state S, two unmodified rollouts produce recipient and donor futures F_R, F_D, actions $\mathbf{a}_R, \mathbf{a}_D$, and one-chunk endpoints $\mathbf{e}_R, \mathbf{e}_D$ (Figure 1). We restore S and recompute the recipient while replacing only its future representation with F_D. Observation, instruction, recipient random draws, solver, denoising schedule, and action coordinates remain fixed. The donor target is reimposed at every denoising step. Cosmos Policy uses an encoding of proprioception and two camera views at the donor's executed endpoint; Cosmos 3 uses the donor's predicted video latent or an encoded video of its executed action. Appendix A.2 gives the exact mechanics.

For action or endpoint $\mathbf{x} \in \{\mathbf{a}, \mathbf{e}\}$, normalized donor projection is

$$\pi_x(\hat{\mathbf{x}}; R, D) = \frac{(\hat{\mathbf{x}} - \mathbf{x}_R)^\top (\mathbf{x}_D - \mathbf{x}_R)}{\|\mathbf{x}_D - \mathbf{x}_R\|_2^2}. \tag{1}$$

The recipient is 0 and donor is 1; changes orthogonal to the donor direction receive little weight. Steering subtracts the model-specific self reference. Every endpoint arm is executed after restoring the same state.

We evaluate `Cosmos-Policy-LIBERO-Predict2-2B` on ten LIBERO-Long tasks at the first query and after three and six open-loop chunks (30 registered states), and the 16B `Cosmos3-Nano-Policy-DROID` checkpoint on six RoboLab tasks (24 frozen state clusters; 22 completed) [8, 11, 14–16, 19]. Pairs are selected by native action and endpoint separation, never intervention outcomes. Where applicable, self, matched-Gaussian, natural-alternative, shuffled-future, and bitwise no-op controls test recomputation, perturbation size, and generic replacement. State is the independent unit; repeated draws and reciprocal directions are averaged within state, with 10,000 state-bootstrap resamples. Full designs appear in Appendix A.1–A.6.

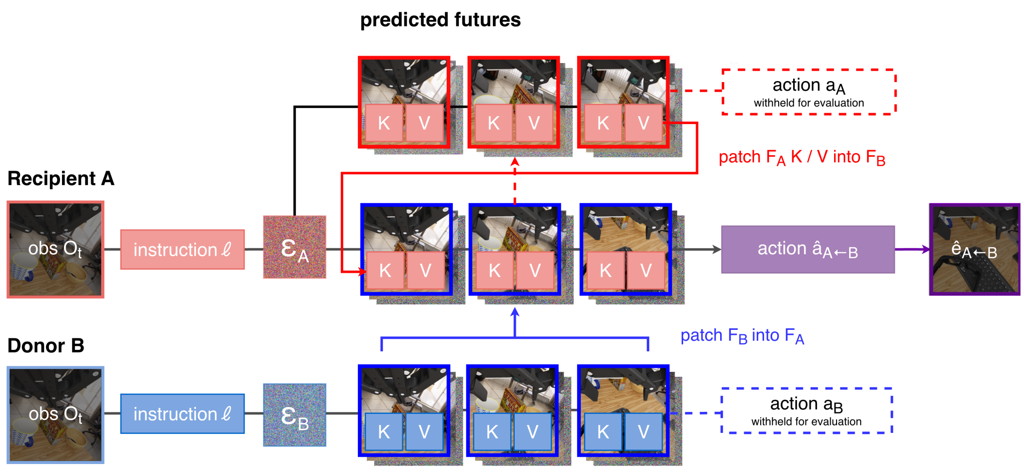

Figure 2: Stage-2 pathway intervention. The donor future remains inserted in recipient A, but predicted-future-position K/V is restored from A's native run. The decrease in donor-directed steering measures how much of the Stage-1 effect passes through this interface.

2.2 Stage 2: K/V pathway intervention

Stage 2 tests model-specific future-to-action attention routes. In Cosmos 3, we record K/V at predicted-future positions in the recipient's native run, leave F_D inserted, and restore only those K/V states. The decrease in donor steering is *K/V replacement loss*. Cosmos Policy removes future-position keys at a prespecified block.

For Cosmos 3, recipient K/V is recorded across all 36 layers and eight model forwards and replayed call by call during the donor-future run. Only action-query rows are recomputed; all other positions and states remain native to that run. The interface was selected on excluded calibration tasks and frozen before population testing. Cosmos Policy gates keys at predicted-future positions in prespecified block 27. Cache alignment, identity controls, calibration, and implementation details appear in Appendix A.3.

Table 1: Stage-1 donor projections. Zero is the self reference and one is the native recipient-to-donor displacement. Intervals and state-level estimates appear in Appendix Figure 3.

Model	Outcome/source	Gaussian	Natural	Shuffled	Paired donor
Cosmos 3	Predicted action	—	0.559	—	**0.992**
	Executed-video action	0.041	0.591	—	**0.983**
	Executed endpoint	-0.041	0.514	—	**1.003**
Cosmos Policy	Action	0.001	0.395	0.244	**0.499**
	Executed endpoint	0.003	0.442	0.275	**0.552**

3 Results

3.1 Coherent futures directionally steer behavior

Across 22 Cosmos 3 states and all six tasks, the paired donor's predicted future produces action steering of 0.992 ($[0.972, 1.010]$; 22/22 states positive), versus 0.559 for a naturally predicted alternative (Table 1). Executed donor videos produce action steering of 0.983, compared with 0.591 for the naturally predicted alternative and 0.041 for the matched-Gaussian control. In secondary analyses, executing the recomputed action yields endpoint steering of 1.003 ($[0.950, 1.050]$; 22/22), versus 0.514 and -0.041 for those controls. Thus, donor-directed change survives execution and is not explained by recomputation, an equally large unstructured displacement, or merely supplying a different valid prediction. Projection measures agreement along the donor direction, not equality of complete vectors.

At Cosmos Policy's first query, the paired donor produces action steering of 0.499 and endpoint steering of 0.552, positive in all ten tasks. Natural controls reach 0.395 and 0.442, shuffled controls 0.244 and 0.275, and Gaussian controls remain near zero. The paired donor exceeds the natural alternative by 0.103 ($[0.048, 0.161]$; 8/10) for actions and 0.110 ($[0.054, 0.166]$; 8/10) for endpoints. Steering is near zero at separately sampled later states, establishing state dependence rather than causal temporal decay.

3.2 Content profiles differ across architectures

Cosmos Policy's early-state effect is concentrated in future visual inputs, especially the wrist-camera target. Wrist-camera replacement produces action steering of 0.435 ($[0.284, 0.605]$), primary-camera replacement 0.101 ($[0.057, 0.144]$), and future-proprioception replacement 0.001 ($[-0.004, 0.008]$). In a separate 20-state rendered-video study, the object-state main effect is approximately zero for action and endpoint, with decoder checks confirming that the imposed distinction remains visible. Across four exploratory natural-pair comparisons, robot-motion futures steer more strongly than object-motion futures, consistent with a role for camera-visible robot motion.

Cosmos 3 instead requires more than either isolated pixel factor. Across ten frozen factor-study states spanning all six tasks, a complete executed donor video strongly steers actions (0.988) and endpoints (0.929), but robot-pixel effects are 0.002 and 0.010 and object-pixel effects 0.006 and approximately zero. Every isolated-factor interval lies within the prespecified $[-0.10, 0.10]$ equivalence region. The effect may depend on distributed features or temporal and robot–object coherence. The result establishes factor insufficiency, not the irrelevance of robot or object information (Appendix Figure 4).

3.3 Predicted-future K/V carries most Cosmos 3 steering

Replacing K/V at predicted-future positions removes most of the Stage-1 effect (Table 2). Predicted-action steering falls from 0.992 to 0.115, a loss of 0.877 ($[0.842, 0.910]$), or 88%. Executed-action steering falls from 0.983 to 0.169, a loss of 0.814 ($[0.723, 0.900]$), or 83%. Executed-endpoint steering falls from 1.003 to 0.152, a loss of 0.851 ($[0.766, 0.922]$), or 85%. Loss is positive in all 22 states. Because the donor future remains inserted while positions and non-targeted states stay fixed, this identifies predicted-future K/V as a pathway carrying donor-specific influence. Nonzero residuals make the pathway influential but incomplete.

Table 2: Cosmos 3 K/V pathway test across 22 states. Replacement loss is the within-state decrease in donor steering; intervals are 95% state-bootstrap intervals.

Outcome	Before	After	Loss [95% CI]
Predicted action	0.992	0.115	0.877 [0.842, 0.910]
Executed action	0.983	0.169	0.814 [0.723, 0.900]
Executed endpoint	1.003	0.152	0.851 [0.766, 0.922]

In Cosmos Policy, progressive key removal at block 27 produces a monotonic action-disruption curve. The absolute mean executed-endpoint projection change is 0.281 ($[0.164, 0.430]$; 10/10 states), exceeding equal-count random-key removal by 0.169 ($[0.058, 0.322]$; 9/10). This demonstrates sensitivity to future-position keys at block 27; current-key removal produces a statistically indistinguishable change.

4 Discussion and Limitations

Controlled interventions turn an ambiguous co-generated output into testable claims about what changes behavior and how that effect is transmitted. Across both WAMs, the inserted future's identity predicts the direction of the resulting action and simulator endpoint. Cosmos Policy's early-state effect is concentrated in visual future targets, especially the wrist camera, whereas Cosmos 3's complete-video effect is not recreated by isolated robot or object pixels. Pathway tests show Cosmos Policy's sensitivity to future-position keys at block 27 and that K/V at predicted-future positions carries most of Cosmos 3's donor-directed shift.

This evidence remains narrower than planning. Planning would require generating alternatives, predicting their consequences, comparing them using a goal or value, and selecting among them. We do not test that sequence or establish stable success/failure mediation. The results instead show causal conditioning on coherent future content and identify one important internal route by which that content influences action.

Limitations. The main experiments evaluate two checkpoints in simulation across six RoboLab tasks and LIBERO-Long. They establish directional effects on actions and one-chunk endpoints rather than closed-loop task success. Pixel masks test localized visual factors, and the pathway interventions isolate specified attention interfaces rather than the entire circuit. Broader checkpoints and physical robots remain to be tested.

Responsible use. These methods could improve auditing by distinguishing visually plausible predictions from representations that actually control behavior. They could also redirect robot actions or encourage false confidence on narrow state distributions. Before physical deployment, mitigations should include broader coverage, independent replication, physical safety constraints, human oversight, and monitoring beyond the intervention metric.

5 Conclusion

Together, these interventions provide a causal framework for testing how specific imagined futures direct action and tracing their influence through future-token K/V.

References

[1] Yilun Du, Sherry Yang, Bo Dai, Hanjun Dai, Ofir Nachum, Joshua B. Tenenbaum, Dale Schuurmans, and Pieter Abbeel. Learning universal policies via text-guided video generation. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/1d5b9233ad716a43be5c0d3023cb82d0-Abstract.html.

[2] Yilun Du, Sherry Yang, Pete Florence, Fei Xia, Ayzaan Wahid, Brian Ichter, Pierre Sermanet, Tianhe Yu, Pieter Abbeel, Joshua B. Tenenbaum, Leslie Kaelbling, Andy Zeng, and Jonathan Tompson. Video language planning. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=9pKtcJcMP3.

[3] Frederik Ebert, Chelsea Finn, Sudeep Dasari, Annie Xie, Alex Lee, and Sergey Levine. Visual foresight: Model-based deep reinforcement learning for vision-based robotic control. *arXiv preprint arXiv:1812.00568*, 2018. URL https://arxiv.org/abs/1812.00568.

[4] C. A. Field and A. H. Welsh. Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B*, 69(3):369–390, 2007. doi: 10.1111/j.1467-9868.2007.00593.x.

[5] Nicholas Goldowsky-Dill, Chris MacLeod, Lucas Sato, and Aryaman Arora. Localizing model behavior with path patching, 2023. URL https://arxiv.org/abs/2304.05969. arXiv preprint, version 2.

[6] Yanjiang Guo, Yucheng Hu, Jianke Zhang, Yen-Jen Wang, Xiaoyu Chen, Chaochao Lu, and Jianyu Chen. Prediction with action: Visual policy learning via joint denoising process. In *Advances in Neural Information Processing Systems*, volume 37, pages 112386–112410. Curran Associates, Inc., 2024. doi: 10.52202/079017-3570. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/cbe25fa0e7c7084049276888a09acc8d-Abstract-Conference.html.

[7] Yucheng Hu, Yanjiang Guo, Pengchao Wang, Xiaoyu Chen, Yen-Jen Wang, Jianke Zhang, Koushil Sreenath, Chaochao Lu, and Jianyu Chen. Video prediction policy: A generalist robot policy with predictive visual representations. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 24328–24346. PMLR, 2025. URL https://proceedings.mlr.press/v267/hu25g.html.

[8] Moo Jin Kim, Yihuai Gao, Tsung-Yi Lin, Yen-Chen Lin, Yunhao Ge, Grace Lam, Percy Liang, Shuran Song, Ming-Yu Liu, Chelsea Finn, and Jinwei Gu. Cosmos Policy: Fine-tuning video models for visuomotor control and planning. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=wPEIStHxYH.

[9] Daniël Lakens. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362, 2017. doi: 10.1177/1948550617697177.

[10] Shuang Li, Yihuai Gao, Dorsa Sadigh, and Shuran Song. Unified video action model, 2025. URL https://arxiv.org/abs/2503.00200. arXiv preprint, version 3.

[11] Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking knowledge transfer for lifelong robot learning. In *Advances in Neural Information Processing Systems*, volume 36, pages 44776–44791. Curran Associates, Inc., 2023. doi: 10.52202/075280-1939. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/8c3c666820ea055a77726d66fc7d447f-Abstract-Datasets_and_Benchmarks.html.

[12] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372. Curran Associates, Inc., 2022. doi: 10.52202/068431-1262. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/6f1d43d5a82a37e89b0665b33bf3a182-Abstract-Conference.html.

[13] Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. RoboCasa: Large-scale simulation of household tasks for generalist robots. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, July 2024. doi: 10.15607/RSS.2024.XX.050. URL https://www.roboticsproceedings.org/rss20/p050.html.

[14] NVIDIA. Cosmos 3: Omnimodal world models for physical AI. *arXiv preprint arXiv:2606.02800*, 2026. doi: 10.48550/arXiv.2606.02800. URL https://research.nvidia.com/labs/cosmos-lab/cosmos3/technical-report.pdf.

[15] NVIDIA. Cosmos3-nano-policy-droid. Hugging Face model checkpoint and configuration, 2026. URL https://huggingface.co/nvidia/Cosmos3-Nano-Policy-DROID. Checkpoint revision 2b9f9517efcfbf26e222945b386ae9b65c0930ac.

[16] NVIDIA. Cosmos-policy-libero-predict2-2b. Hugging Face model checkpoint and configuration, 2026. URL https://huggingface.co/nvidia/Cosmos-Policy-LIBERO-Predict2-2B. Checkpoint revision cb689ec0e3347c13667d70a78a3447388f5c3bb8.

[17] Lu Qiu, Yizhuo Li, Yi Chen, Yuying Ge, Yixiao Ge, and Xihui Liu. Making foresight actionable: Repurposing representation alignment in world action models, 2026. URL https://arxiv.org/abs/2606.12217. arXiv preprint.

[18] Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33, pages 12388–12401. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper/2020/hash/92650b2e92217715fe312e6fa7b90d82-Abstract.html.

[19] Xuning Yang, Rishit Dagli, Alex Zook, Hugo Hadfield, Ankit Goyal, Stan Birchfield, Fabio Ramos, and Jonathan Tremblay. RoboLab: A high-fidelity simulation benchmark for analysis of task generalist policies. In *Proceedings of Robotics: Science and Systems*, Sydney, Australia, July 2026. doi: 10.15607/RSS.2026.XXII.096. URL https://roboticsproceedings.org/rss22/p096.html.

[20] Seonghyeon Ye, Yunhao Ge, Kaiyuan Zheng, Shenyuan Gao, Sihyun Yu, George Kurian, Suneel Indupuru, You Liang Tan, Chuning Zhu, Jiannan Xiang, Ayaan Malik, Kyungmin Lee, William Liang, Nadun Ranawaka, Jiasheng Gu, Yinzhen Xu, Guanzhi Wang, Fengyuan Hu, Avnish Narayan, Johan Bjorck, Jing Wang, Gwanghyun Kim, Dantong Niu, Ruijie Zheng, Yuqi Xie, Jimmy Wu, Qi Wang, Ryan Julian, Danfei Xu, Yilun Du, Yevgen Chebotar, Scott Reed, Jan Kautz, Yuke Zhu, Linxi “Jim” Fan, and Joel Jang. World action models are zero-shot policies, 2026. URL https://arxiv.org/abs/2602.15922. arXiv preprint.

[21] Tianyuan Yuan, Zibin Dong, Yicheng Liu, and Hang Zhao. Fast-WAM: Do world action models need test-time future imagination?, 2026. URL https://arxiv.org/abs/2603.16666. arXiv preprint, version 2.

[22] Chushan Zhang, Jinguang Tong, Xuesong Li, Yikai Wang, and Hongdong Li. Keep the future, drop the rollout: RIFT for world action models, 2026. URL https://arxiv.org/abs/2608.11521. arXiv preprint, version 2.

[23] Fred Zhang and Neel Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=Hf17y6u9BC.

A Reproducibility and Protocol Details

A.1 Experimental settings and frozen designs

The Cosmos Policy experiments use the public `Cosmos-Policy-LIBERO-Predict2-2B` checkpoint [16], LIBERO-Long, PyTorch 2.7 with CUDA 12.8, and two NVIDIA H100 80GB GPUs. The frozen design contains ten tasks, three state indices, eight native policy runs per state, one reciprocal donor pair per state, and three predicted-future noise draws per direction. The observation prefix, instruction, recipient action noise, predicted-future noise schedule, tokenizer settings, and denoising schedule are held fixed within an intervention comparison.

The Cosmos 3 experiments use the public 16B `Cosmos3-Nano-Policy-DROID` checkpoint with the released Cosmos, Cosmos Framework, and RoboLab evaluation stack. The frozen population screen selected 24 state clusters across six eligible tasks, capped at four clusters per task, using only native success, donor separation, and replay-validity information. Twenty-two clusters completed. The separate robot/object factor study selected 12 clusters by native object-position separation and completed its prespecified minimum of ten across all six tasks.

A.2 How the donor future is inserted into the recipient run

Cosmos Policy. The donor target is a clean latent encoded with the public tokenizer from the RGB and proprioceptive endpoint reached by a normal 16-action rollout. In the released latent layout, temporal indices 5, 6, and 7 are predicted future proprioception, wrist RGB, and primary RGB. At initialization and at each of the five solver evaluations ($\sigma = 80.0, 42.291, 20.972, 9.618, 4.0$), we replace only these recipient predicted-future coordinates with the donor values. The denoiser input at noise level σ is $F_D + \sigma\epsilon_R$, where ϵ_R is the recipient's fixed predicted-future noise draw, and the returned clean-x_0 estimate is reset to F_D at the same coordinates. The action and value coordinates are never written. Thus, the donor target is reimposed at every solver evaluation rather than inserted once into a noisy state or predicted velocity. The evaluated direct-policy configuration is always conditional (its unconditioned input is `None`), so every solver call uses the wrapped x_0 function.

Cosmos 3. The prepared rectified-flow state is flattened as vision then action: vision has shape $[1, 48, 9, 33, 40]$ and action $[33, 64]$. We preserve latent frame 0 and replace predicted-future video frames 1–8 with the donor target throughout denoising. A target F_D is either the final clean video latent from a normal prediction or the public-tokenizer encoding of a 33-frame video produced by executing a normal 32-action rollout. Before both model forwards at each of four UniPC calls, predicted-future vision is set to $(1-\sigma)F_D+\sigma\epsilon_R$. After guidance-3 combines the predicted velocities, we replace only the predicted-future-vision velocity by $(x_\sigma - F_D)/\sigma$, making its clean estimate F_D at the sampler boundary. The recorded sigmas are $0.999, 0.937, 0.833, 0.624$. The current-video and action state and action velocity are never changed.

A.3 K/V pathway implementation

The Cosmos 3 run with the recipient's own predicted future records keys and values at predicted-future video positions separately for all 36 layers and all eight model forwards: four denoising calls, each with conditional and unconditional passes. Cache entries are replayed call by call in the run with the donor's predicted future, and only action-query rows are recomputed. Query outputs at current-video and predicted-future-video positions remain native; text, current-video, and action K/V are unchanged. A calibration scan on excluded tasks selected layer 16 for single-layer localization, whereas the population estimate replaces all 36 direct K/V interfaces from predicted-future positions to action queries.

A.4 State restoration, execution, and missingness

Each endpoint comparison begins from an identical saved simulator state. Evaluation reconstructs the environment, restores that state, replays the observation prefix, verifies the restored observation, and then executes one recomputed action chunk. Recipient and donor endpoints are produced by the same restoration and execution procedure. This design makes the endpoint projection a comparison between reachable continuations of one physical state rather than between unrelated trajectories.

Two frozen Cosmos 3 population units were unavailable because their registered source branches ended after 21 and 29 actions, before producing the required 32-action, 33-frame intervention source. Neither unit was replaced, and no intervention outcome entered the decision to retain or omit a state. The completed population therefore exceeds the prespecified minimum of 20 states across at least five tasks. The factor study likewise meets its minimum of ten states across at least five tasks.

A.5 Intervention validity and controls

For Cosmos Policy, the self condition passes the recipient target through the complete intervention machinery. For Cosmos 3, the primary self reference is an exact native recomputation. The mechanics-matched clean self-clamp has mean donor projection -0.006; donor-minus-self-clamp steering remains 0.999 and is positive in 22/22 states. Identity token-layout capture and K/V record/replay are bitwise exact relative to the same clamped-self trajectory. Sentinels at the sampler boundary confirm that replacing recipient predicted-future coordinates with donor values does not write directly into action coordinates. The pathway experiment at predicted-future token positions preserves token count and order while leaving current-video, language, action, and all non-targeted key/value states unchanged.

The geometry-matched Gaussian control has the same norm and recipient distance as the donor displacement but no coherent content. Natural controls are valid unpaired alternative future targets selected without intervention outcomes, and shuffled controls use another target from the same task; the Cosmos 3 natural target is a model-predicted future. These comparisons distinguish exact recomputation, perturbation magnitude, effects of generic alternative targets, and pair-specific content. Executed-video conditions add a separate realizability check by encoding frames generated when a native donor action is physically replayed in the simulator.

A.6 Stage-1 statistical analysis

The independent unit is a restored simulator state, not an individual noise draw or recipient–donor direction. Cosmos Policy reciprocal directions and repeated noise draws are averaged within state; each Cosmos 3 state uses one frozen recipient–donor orientation and recipient diffusion path. Reported 95% intervals use 10,000 percentile-bootstrap resamples of states [4]. For the common-scale condition estimates reported in Section 3.1, each control's projection is reconstructed within unit from the registered pairwise contrasts before bootstrapping; these are not differences of aggregate means. Task-balanced sensitivity analyses first average states within task and then resample tasks.

We prespecified 0.10 projection units—10% of the native recipient-to-donor displacement—as the smallest effect of interest for the Cosmos Policy confirmatory family and the Cosmos 3 population and factor families [9]. An interval contained within $[-0.10, 0.10]$ supports practical equivalence to zero at that scale. Prespecified primary sign tests use state-level directions and Holm correction within each family.

A.7 Stage-2 statistical analysis

Stage 2 reuses the Stage-1 independent-unit and bootstrap conventions. For Cosmos 3, replacement loss is formed within restored state by subtracting K/V-patched donor steering from the corresponding unpatched donor steering before population aggregation. Task-balanced sensitivity estimates and family-wise Holm correction follow the Stage-1 procedure. The Cosmos Policy key-removal study uses ten first-query states; its endpoint intervals are reported in Appendix B.5.

B Additional Experiments and Diagnostics

B.1 Stage-1 directional steering

Figure 3 displays the Stage-1 estimates at the state or task level. Panel a shows donor-minus-self steering, while panel b shows the paired donor's advantage over each control. The unit-level markers expose variation suppressed by the aggregate estimates reported in the main text.

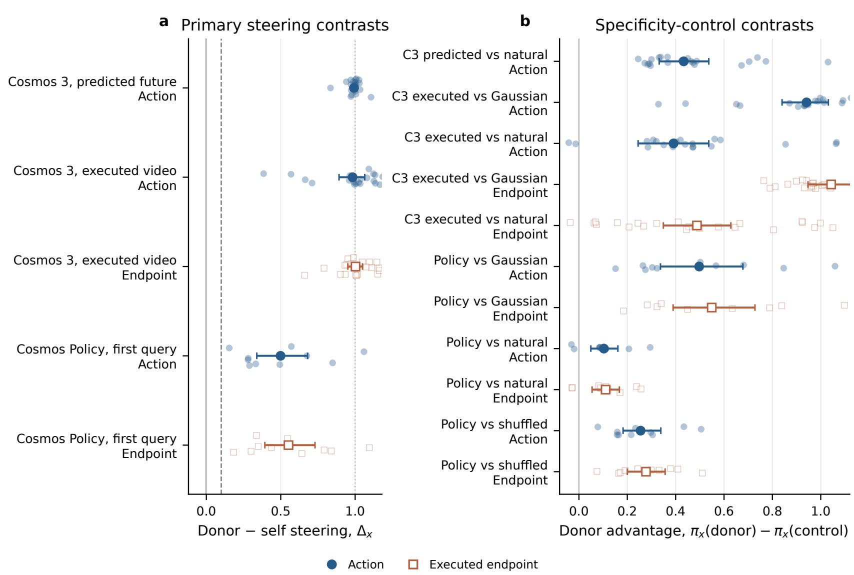

Figure 3: Stage-1 directional steering (a) and donor-minus-control contrasts (b). Small markers are states for Cosmos 3 and tasks for Cosmos Policy; large markers and whiskers are means and 95% bootstrap intervals. Blue denotes actions and orange denotes endpoints.

B.2 Cosmos 3 robot/object factorization diagnostic

Figure 4 shows the full 2×2 hybrid intervention summarized in Section 3.2. Panels a and b retain the four pixel-source cells and state-level estimates; panel c places the two factorial main effects beside the full executed-donor-video benchmark.

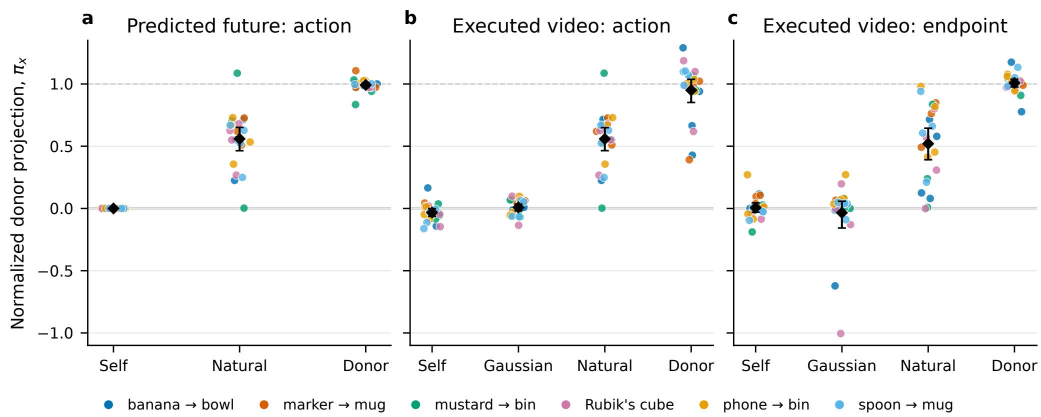

Figure 5: Cosmos 3 condition-level projections across 22 restored states for predicted actions, executed actions, and endpoints. Colors identify tasks; black diamonds and whiskers are means and 95% state-bootstrap intervals.

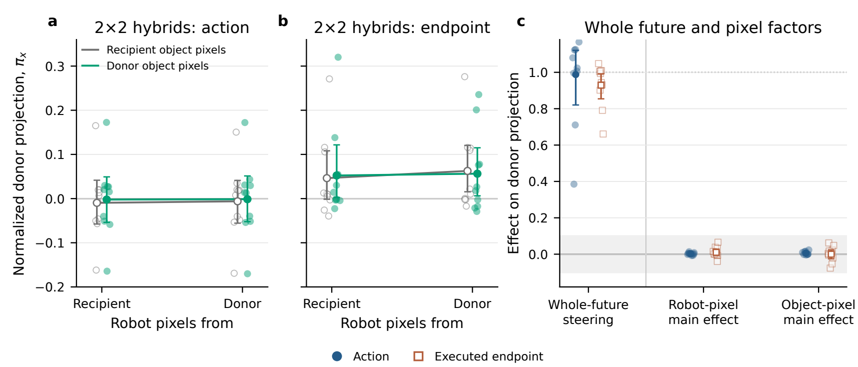

Figure 4: Cosmos 3 robot/object pixel factorization. Panels (a)–(b) show the four hybrid-video cells; panel (c) compares robot/object main effects with the full executed-donor-video benchmark. Small markers are states; large markers and whiskers are means and 95% bootstrap intervals.

B.3 Condition-level Cosmos 3 projections

Figure 5 shows the condition estimates before they are converted into donor-minus-control contrasts. Recomputation with the recipient's own predicted future stays at zero, naturally predicted alternatives produce partial donor-directed movement, and paired donors cluster near one. For executed videos, the matched Gaussian remains near zero while both action and endpoint effects preserve the same ordering. The state-level display also makes clear that the endpoint conclusion is not produced by a single task or a single extreme unit.

B.4 Displacement-magnitude diagnostic

Directional projection could be misleading if a large normalized estimate arose from an extremely small native denominator or if inserting the donor's predicted future caused only a negligible displacement. Figure 6 therefore compares unnormalized recipient-to-donor separation with the displacement caused by replacing the recipient's predicted future with the donor's. Predicted-action and executed-endpoint displacements lie close to the identity line across the observed range; executed-action displacement is more variable but remains on the scale of the native difference. This diagnostic supports the interpretation that the intervention recreates a substantial behavioral displacement, not merely a tiny change that happens to align with the donor axis.

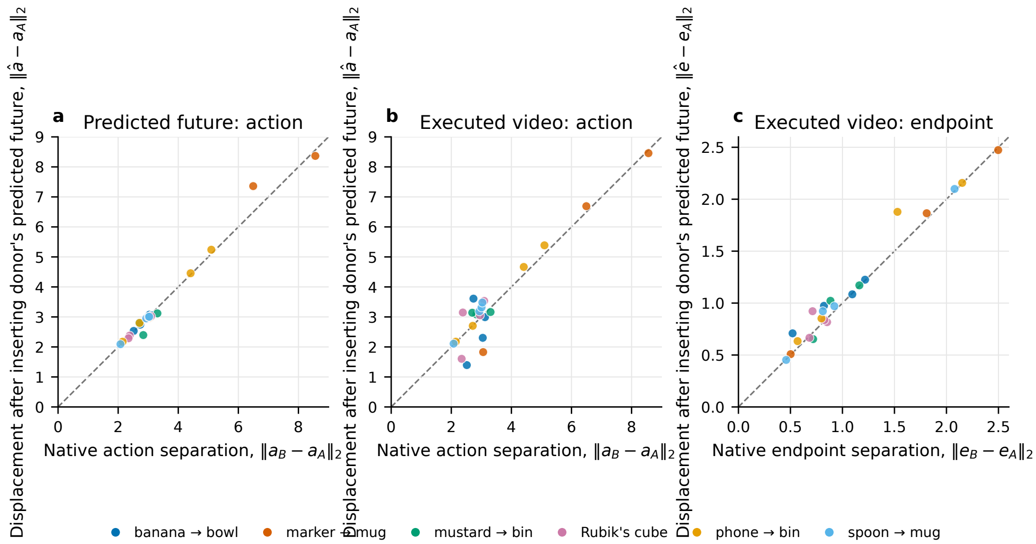

Figure 6: Native recipient-to-donor separation versus predicted-future-insertion displacement for Cosmos 3 predicted actions, executed actions, and endpoints. Dashed lines indicate equality; colors identify tasks.

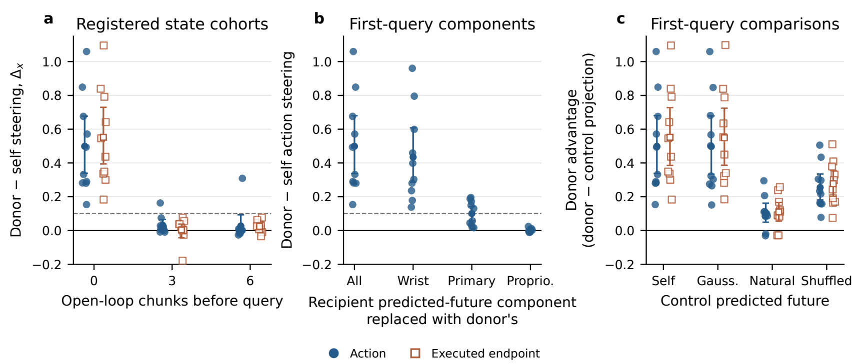

Figure 7: Cosmos Policy scope checks: (a) state cohorts, (b) future-target components, and (c) donor-minus-control contrasts. Small markers are tasks; large markers and whiskers are means and 95% task-bootstrap intervals.

B.5 Cosmos Policy timing and cross-domain checks

Across all 30 registered Cosmos Policy states, mean donor-minus-self steering is 0.186 ($[0.095, 0.290]$) for actions and 0.194 ($[0.091, 0.305]$) for endpoints. The first-query, after-three-chunk, and after-six-chunk action estimates are 0.499, 0.031, and 0.027; the corresponding endpoint estimates are 0.552, 0.004, and 0.025. Because the three indices represent different physical states that may be nested within task trajectories, the first-query analysis is the cleaner confirmatory result and the differences should not be read as causal temporal decay.

The six-state RoboCasa replication is exploratory and is not pooled with LIBERO [13]. Donor-minus-self action steering is 0.0144 ($[0.0034, 0.0276]$), while its donor-minus-Gaussian contrast is 0.0025 with an interval crossing zero. Endpoint steering is 0.0287 ($[0.0101, 0.0492]$), positive in 6/6 states; donor-minus-Gaussian endpoint steering is 0.0187 ($[0.0063, 0.0329]$), also positive in 6/6.

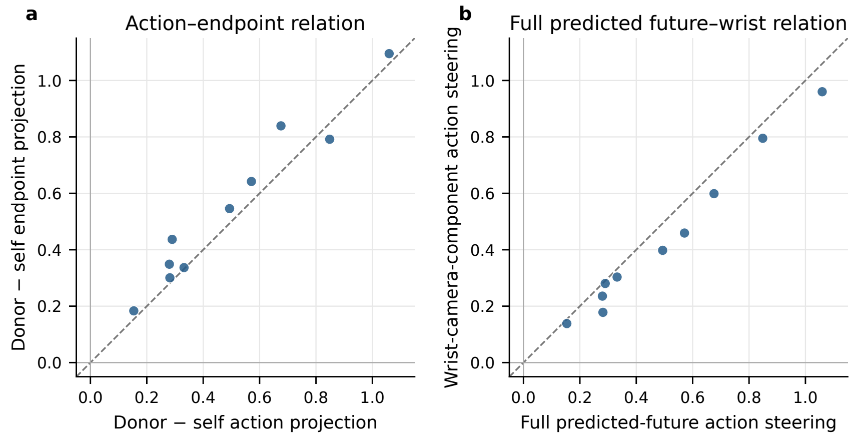

Figure 8: Cosmos Policy task-level relationships at the first query: (a) action versus endpoint steering and (b) full-target versus wrist-camera-component steering. Points are tasks; dashed lines indicate equality.

B.6 Within-task relationships in Cosmos Policy

Figure 8 provides two task-level diagnostics at the first query. Action and executed-endpoint steering generally rise together, supporting the use of simulator execution as a behavioral continuation of the action result. Steering from the wrist-camera component also tracks steering from the full future target across tasks, consistent with the aggregate modality decomposition while showing that the wrist view does not reproduce the entire effect uniformly.

B.7 Additional content and pathway checks

The Cosmos Policy rendered 2×2 study supplies a manipulation check for the object-content null. Decoded future targets identify the intended hybrid cell with 92.5% accuracy in the primary camera and 98.75% in the wrist camera, while executed endpoints identify the four cells at the 25% chance rate. Thus, the future representation visibly contains the imposed distinction even though the preregistered object main effect on action and goal-state endpoint is approximately zero. Four states met both natural-pair criteria; in these comparisons, robot-motion donors steer more strongly than object-motion donors.

The block-27 Cosmos Policy key-removal analysis includes two additional checks beyond the action-space dose curve reported in the main text. Removing all keys at predicted-future positions changes the executed endpoint by an absolute mean projection of 0.281 ($[0.164, 0.430]$; 10/10 first-query states) and exceeds equal-count random-key removal by 0.169 ($[0.058, 0.322]$; 9/10). Its difference from equal-count current-key removal is -0.035 with an interval crossing zero. The endpoint analysis therefore supports an active route from the predicted future at the first query while reinforcing the conclusion that this route is not uniquely dominant over current-state information.

For Cosmos 3, task-balanced pathway-loss estimates are 0.877 for predicted actions, 0.812 for executed-video actions, and 0.853 for executed endpoints, closely matching the state-weighted estimates. The remaining steering after K/V replacement using the recipient's own predicted future is 0.115, 0.169, and 0.152, respectively. This residual motivates finer-grained tests of parallel attention heads, non-attention routes, and interactions between patched and unpatched components.

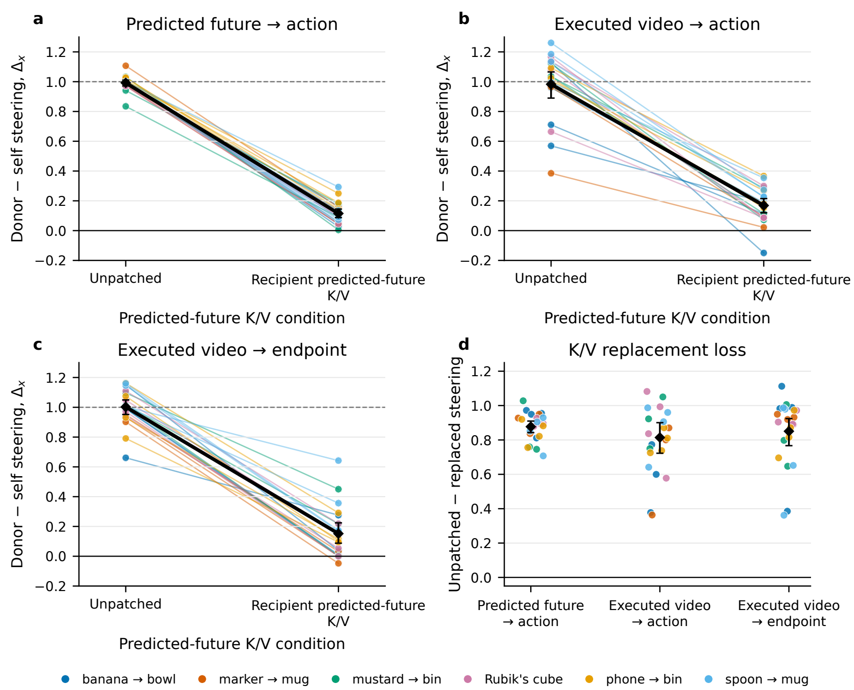

Figure 9: Cosmos 3 donor steering before and after K/V replacement at predicted-future token positions. Lines connect the same state; black diamonds and whiskers are means and 95% state-bootstrap intervals. Panel (d) shows replacement loss.